INVESTIGATING EARTH ORIENTATION PARAMETERS AS PREDICTORS OF GLOBAL TEMPERATURE ANOMALIES

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ABSTRACT

Earth's rotation is not constant, it varies due to geophysical and astronomical forces, which are recorded as Earth Orientation Parameters (EOPs). These parameters include polar motion and changes in the length of day and are well-documented by the international Earth Rotation Service. This project investigates whether changes in EOp dan be used to model or predict global temperature anomalies over decadal time scales.

To test this idea, synthetic data was created to simulate 600 months of EOP and global temperature anomaly records using R. Each dataset included realistic noise, linear drift, and periodic components to reflect underling geophysical variability. A 24-month moving average was applied to both time series o extract low-frequency trends, following methods comparable to a single iteration of the Kolmogorov–Zurbenko (KZ) filter. The relationship between EOP and temperature was assessed using Pearson correlation and linear regression.

Results showed a statistically significant moderate positive correlation between the two smoothed signals, with a Pearson r value of 0.584 and a p-value less than 3.17 x 10^{-58}, which means the relationship is statistically significant and very unlikely to be random.

This study supports the hypothesis that EOP signals can encode climate-relevant information. This method used in this project could be applied to real data from NASA's GISTEMP temperature records and International Earth Rotation Service's EOP data. In the future, this work could be expanded by using wavelet analysis or machine learning models to explore how EOP might improve climate forecasting.

BACKGROUND

- Earth Orientation Parameters (EOPs) measure Earth's rotational behavior, including polar motion and length-of-day changes [1][2].
- EOP data is not commonly used in climate models but may reveal long-term relationships [2][8].
- Some studies show that these signals relate to internal Earth system changes, like temperature variability [3][9].
- This study investigates whether EOP variability correlates with synthetic temperature anomalies.

OBJECTIVES

- Create realistic EOP and temperature anomaly simulations for long-term analysis
- Use a 24-month moving average to extract slow-changing patterns
- Test for statistical correlation using Pearson r and regression
- Validate trend alignment and significance
- Develop a pipeline for real-world datasets (e.g., IERS and NASA GISTEMP)

Can Earth's rotational behavior be used to predict temperature trends over time?

METHODOLOGY

- **Data Generation**: 600-month EOP and temperature datasets simulated using sine waves, linear trends, and noise
- **Signal Smoothing**: Applied 24-month centered moving average (KZ filter, 1 iteration) [11]
- **Statistical Analysis**: Pearson correlation (r = 0.584), linear regression, p-value testing
- **Tools Used**:
  - R (zoo, ggplot2, roll) [16–20]
  - NASA GISTEMP [12], IERS EOP Data [13]

Pearson r = 0.584
p < 3.17 × 10^{-58}
24-month moving average window (1 iteration, KZ filter)

RESULTS

Long-Term Trends in EOP and Temperature

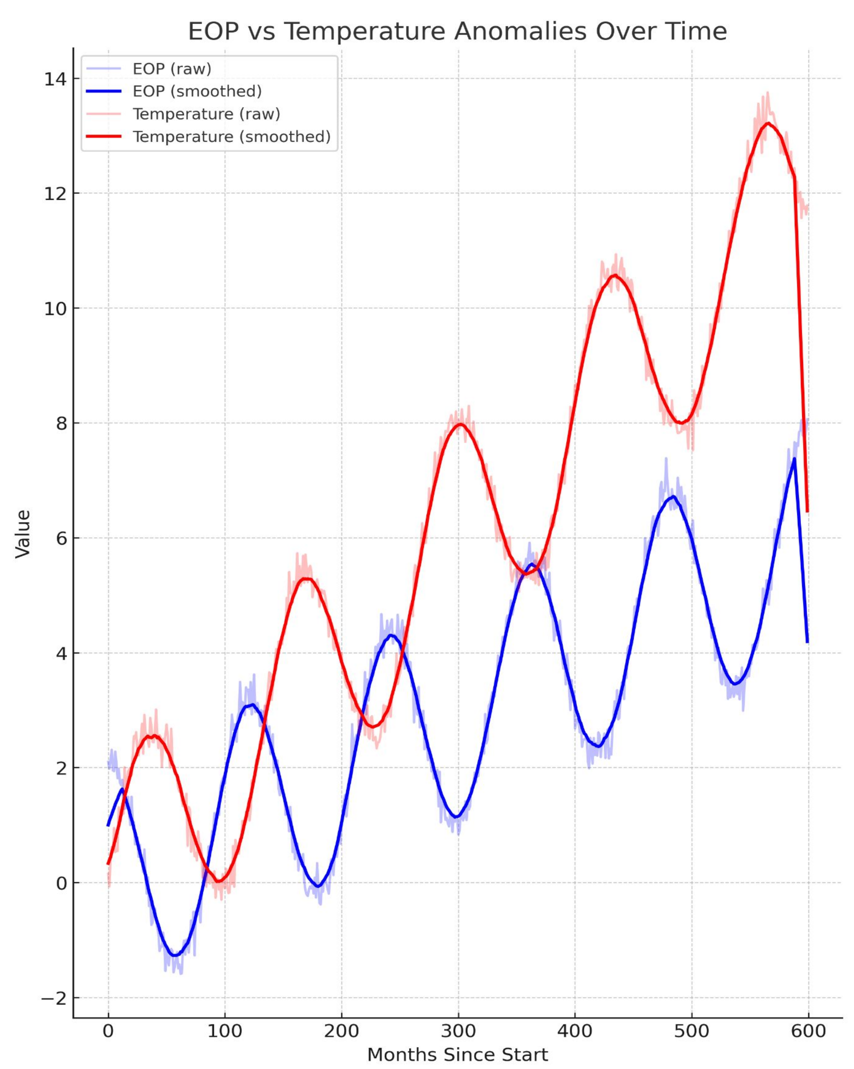

Fig 1. Time Series of simulated Earth Orientation Parameters (EOP) and global temperature anomalies over 600 months. Smoothed EOP (blue) and temperature (red) signals show correlated long-term trends, supporting the hypothesis that changes in Earth's rotation may influence near term temperature behavior.

EOP-Temperature Correlation

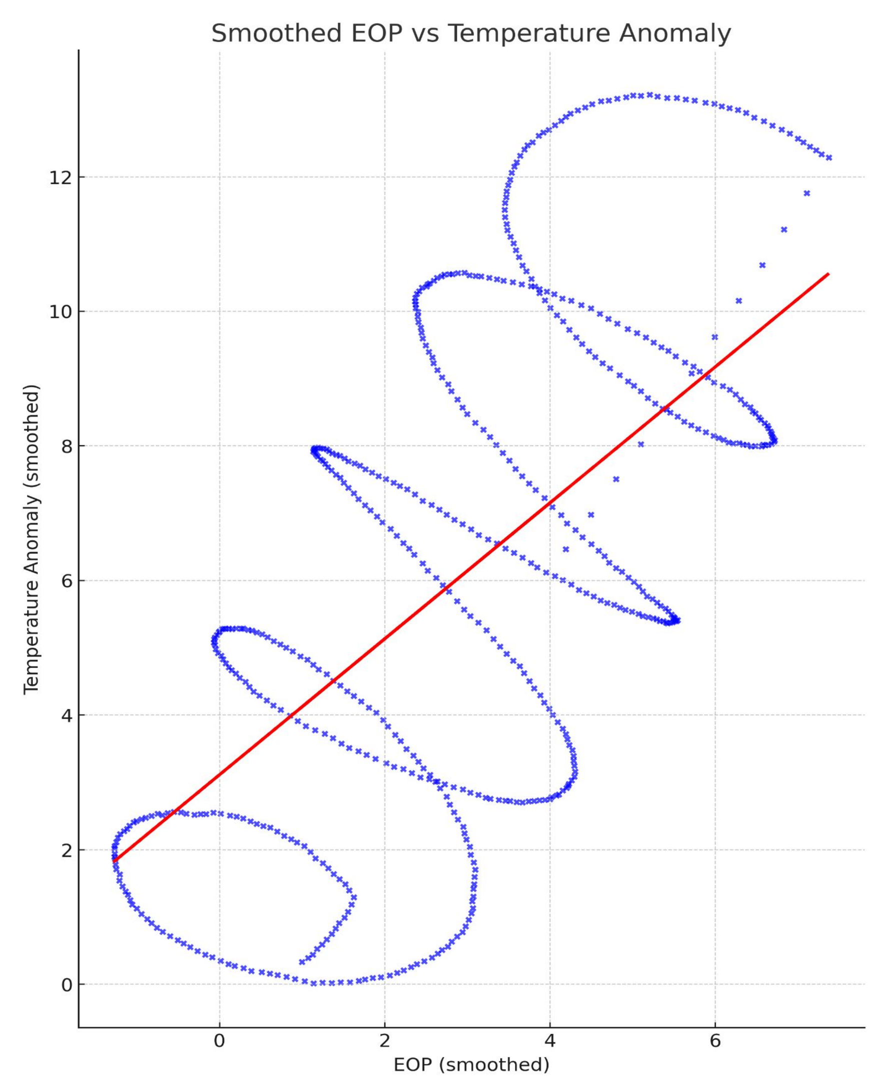

Fig 2. Scatter Plot comparing smoothed EOP and temperature values. The linear regression suggests that EOP variability may serve as a valuable predictor for climate forecasting models based on rotational climate coupling.

Modeled Periodicity in Temperature Signa: Simulated Temperature Anomaly with Seasonal Components

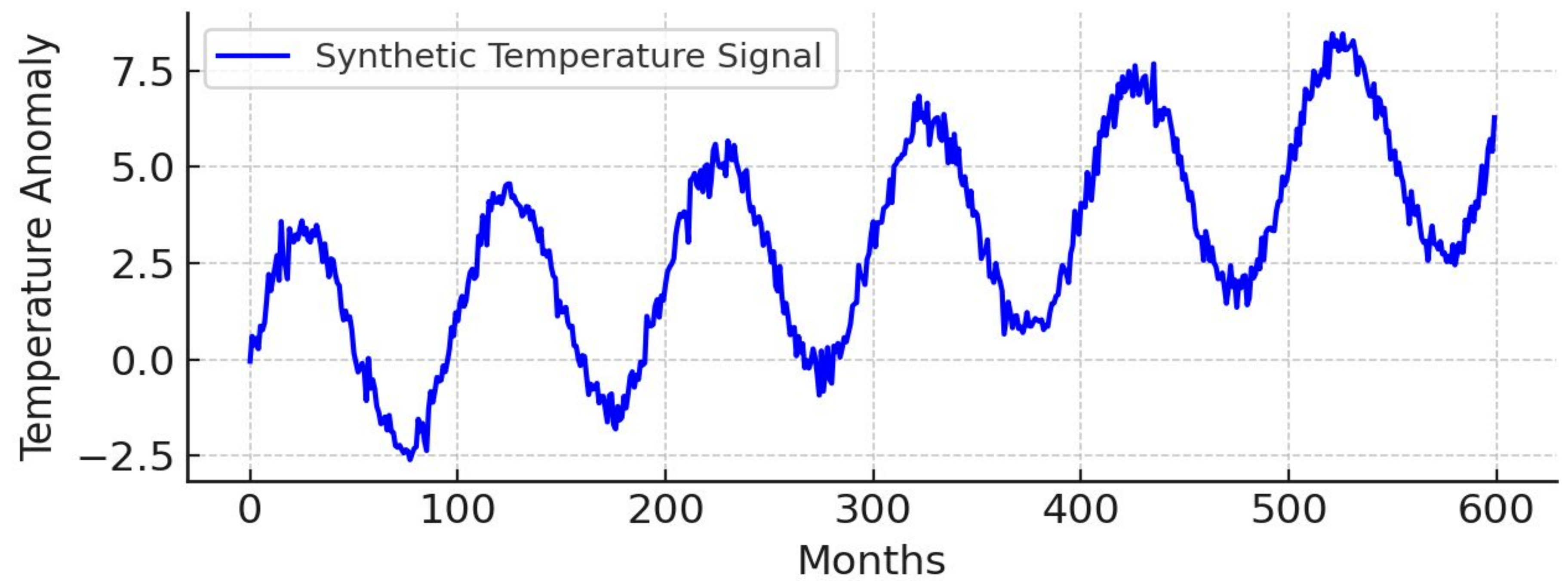

Fig 3. Simulated time series of global temperature anomalies constructed using sinusoidal, trend, and noise components. This figure highlights the synthetic seasonal behavior built into the dataset, helping to mimic natural periodicity such as solar or ENSO-driven cycles.

High-Frequency Noise Isolation: Residuals Between Raw and Smoothed Temperature Signals

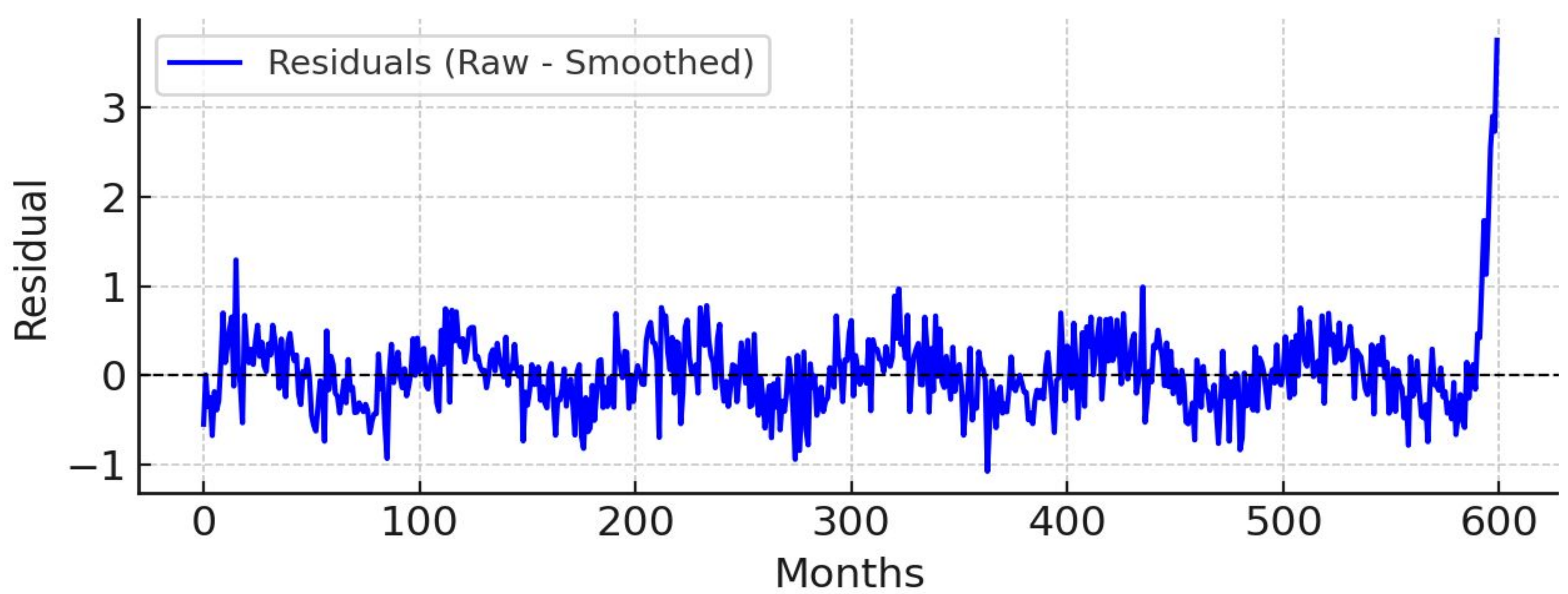

Fig 4. Residuals obtained by subtracting the smoothed signal from the original temperature series. This shows the high-frequency fluctuations removed by the 24-month moving average, supporting the method's ability to isolate long-term temperature patterns relevant to EOP coupling.

DISCUSSION

- Simulated EOP and temperature signals showed a clear statistical relationship after smoothing.
- The correlation supports the hypothesis that rotational dynamics influence climate signals over long time scales [2][3].
- Smoothing techniques such as the 24-month moving average (KZ filter) effectively isolate low-frequency behavior from noise [11].
- These results validate the use of synthetic data to test methods before applying them to real-world climate-EOP datasets.

CONCLUSION AND FUTURE WORK

➔ This study demonstrates that Earth's rotational signals may encode climate-relevant information useful for long-term temperature prediction.

➔ The methodology can be applied to real datasets like NASA GISTEMP [12] and IERS EOP series [13].

➔ Future work will include:

- Using wavelet decomposition to study frequency evolution over time
- Testing machine learning models such as artificial neural networks (ANNs) for forecasting
- Evaluating prediction skill when EOP is added to traditional temperature models [6][14].

These improvements could lead to novel forecasting tools that integrate geophysical signals with atmospheric science.

EOPs may offer a new way to enhance climate prediction models by capturing Earth system dynamics.

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CREDIT AUTHORSHIP AND CONTRIBUTION STATEMENTS

Koyi, A.: simulation design, literature review, RStudio and Python coding, smoothing methods (KZ filter), writing – Abstract, Methods, Results, Discussion, Conclusion, figure creation (Figures 1–4);
Marsellos, A.E.: supervision, spectral analysis guidance, statistical interpretation, tutorial instruction, research planning support, editing;

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